



CAT 2023 Slot 3 Question Paper - QA

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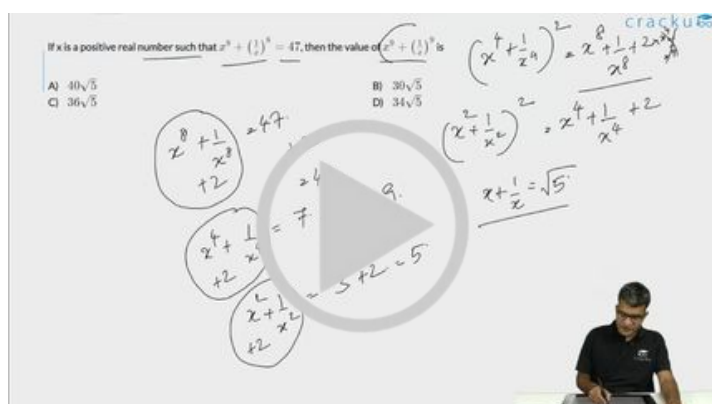
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Quant

1. If x is a positive real number such that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, then the value of $x^9 + \left(\frac{1}{x}\right)^9$ is

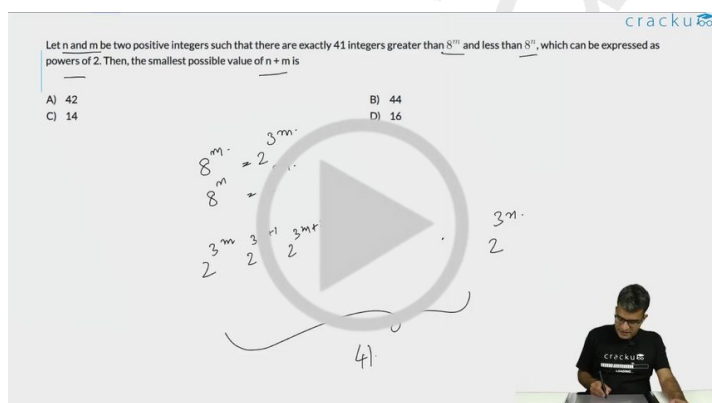
- A $40\sqrt{5}$
- B $30\sqrt{5}$
- C $36\sqrt{5}$
- D $34\sqrt{5}$



VIDEO SOLUTION

2. Let n and m be two positive integers such that there are exactly 41 integers greater than 8^m and less than 8^n , which can be expressed as powers of 2. Then, the smallest possible value of $n + m$ is

- A 42
- B 44
- C 14
- D 16



- A 33
- B 25
- C 15
- D 55

For some real numbers a and b , the system of equations $x + y = 4$ and $(a + 5)x + (b^2 - 15)y = 8b$ has infinitely many solutions for x and y . Then, the maximum possible value of ab is

A) 33 B) 25
C) 15 D) 55

VIDEO SOLUTION

4. For a real number x , if $\frac{1}{2}$, $\frac{\log_3(2^x - 9)}{\log_3 4}$, and $\frac{\log_5(2^x + \frac{17}{2})}{\log_5 4}$ are in an arithmetic progression, then the common difference is

- A $\log_4 \left(\frac{3}{2}\right)$
- B $\log_4 7$
- C $\log_4 \left(\frac{23}{2}\right)$
- D $\log_4 \left(\frac{7}{2}\right)$

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C) $\log_4 \left(\frac{23}{2}\right)$ D) $\log_4 \left(\frac{7}{2}\right)$

VIDEO SOLUTION

5. Let n be any natural number such that $5^{n-1} < 3^{n+1}$. Then, the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$ for each such n , is

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Answer: _____

Handwritten notes:

- $n=1 \rightarrow 1.62 \rightarrow 2.7$
- $n=2 \rightarrow 4.3$
- $n=4 \rightarrow 12$
- $n=5 \rightarrow 15$
- $n=6 \rightarrow 18$
- $\frac{5^n}{5} < 3^{n+1}$
- $\frac{5^n}{5} < 15$
- $3^{n+1} < 2^{n+1}$
- $n=5$
- $1, 2, 3, 4, 5$



VIDEO SOLUTION

6. The sum of the first two natural numbers, each having 15 factors (including 1 and the number itself), is

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Answer: _____

Handwritten notes:

- $N = p_1^{a_1} \times p_2^{a_2} \times p_3^{a_3} \times \dots \times p_n^{a_n}$
- No. of factors = $(a_1+1)(a_2+1) \dots (a_n+1)$
- 1×15
- 15×1
- 3×5
- 5×3
- p^{14}



VIDEO SOLUTION

7. A quadratic equation $x^2 + bx + c = 0$ has two real roots. If the difference between the reciprocals of the roots is $\frac{1}{3}$, and the sum of the reciprocals of the squares of the roots is $\frac{5}{9}$, then the largest possible value of $(b + c)$ is

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Answer: _____

Handwritten notes:

- P, Q
- $x^2 + bx + c = 0$
- $\frac{1}{P} - \frac{1}{Q} = \frac{1}{3}$
- $\frac{1}{P^2} + \frac{1}{Q^2} = \frac{5}{9}$
- $\frac{1}{P} + \frac{1}{Q} = \frac{2}{3}$
- $\frac{2}{PQ} = \frac{4}{9}$
- $PQ = \frac{9}{2}$
- $\frac{1}{P} - \frac{1}{Q} = \frac{1}{3}$
- $\frac{1}{P} + \frac{1}{Q} = \frac{2}{3}$
- $\frac{1}{P} = \frac{1}{3}$
- $\frac{1}{Q} = \frac{1}{3}$
- $P = 3$
- $Q = 3$
- $b = -6$
- $c = 9$
- $b + c = 3$



VIDEO SOLUTION

8. A merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him. He sells the same cloth at a rate of Rs.110 per meter but cheats his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers. If the merchant provides a 5% discount, the resulting profit earned by him is

- A 4.2%
- B 9.7%
- C 15.5%
- D 16%

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A) 4.2%
B) 9.7%
C) 15.5%D) 16%

Handwritten notes on the screen show calculations for the profit percentage, including the purchase price (Rs.100/m), the effective purchase price after the 5% discount (Rs.95/m), and the selling price (Rs.110/m).

VIDEO SOLUTION

9. Rahul, Rakshita and Gurmeet, working together, would have taken more than 7 days to finish a job. On the other hand, Rahul and Gurmeet, working together would have taken less than 15 days to finish the job. However, they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. If Rakshita had worked alone on the job then the number of days she would have taken to finish the job, cannot be

- A 20
- B 17
- C 16
- D 21

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A) 20
B) 17
C) 16
D) 21

Handwritten notes on the screen show the work done by each person and the total work done. The work done by Rahul and Gurmeet is compared to the total work, and the work done by Rakshita is compared to the total work. The calculations show that the total work done is greater than 7(A+B+C) and less than 15(A+B+C).

10. The population of a town in 2020 was 100000. The population decreased by $y\%$ from the year 2020 to 2021, and increased by $x\%$ from the year 2021 to 2022, where x and y are two natural numbers. If population in 2022 was greater than the population in 2020 and the difference between x and y is 10, then the lowest possible population of the town in 2021 was

- A 72000
- B 74000
- C 73000
- D 75000

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A) 72000
B) 74000
C) 73000

Handwritten notes on the video screen:

- 2021: $100000(1-y\%)$
- 2022: $100000(1-y\%)(1+x\%) > 100000$
- Conditions: $x < y$, $x > y$, $y < x$
- Options analysis: $0.9 \times 1.1 < 1$, $0.8 \times 1.2 < 1$, $0.7 \times 1.3 < 1$

11. Anil mixes cocoa with sugar in the ratio 3 : 2 to prepare mixture A, and coffee with sugar in the ratio 7 : 3 to prepare mixture B. He combines mixtures A and B in the ratio 2 : 3 to make a new mixture C. If he mixes C with an equal amount of milk to make a drink, then the percentage of sugar in this drink will be

- A 17
- B 16
- C 21
- D 24

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A) 17
B) 16
C) 21

Handwritten notes on the video screen:

- Mixture A: Cocoa 3, Sugar 2
- Mixture B: Coffee 7, Sugar 3
- Mixture C: 2 parts A, 3 parts B
- Final Drink: Mixture C + equal amount of milk
- Options: A) 17, B) 16, C) 21

12. There are three persons A, B and C in a room. If a person D joins the room, the average weight of the persons in the room reduces by x kg. Instead of D, if person E joins the room, the average weight of the persons in the room increases by $2x$ kg. If the weight of E is 12 kg more than that of D, then the value of x is

- A 2
B 0.5
C 1
D 1.5

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A) 2
B) 0.5
C) 1

$$A+B+C = 3P$$

$$A+B+C+D = 4(P-x) = 4P-4x$$

$$D = P-4x$$

$$A+B+C+D = 4(P-x)$$

$$D = P-4x$$

cracku

cracku

13. A boat takes 2 hours to travel downstream a river from port A to port B, and 3 hours to return to port A. Another boat takes a total of 6 hours to travel from port B to port A and return to port B. If the speeds of the boats and the river are constant, then the time, in hours, taken by the slower boat to travel from port A to port B is

- A $12(\sqrt{5} - 2)$
B $3(3 + \sqrt{5})$
C $3(\sqrt{5} - 1)$
D $3(3 - \sqrt{5})$

cracku

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A) $12(\sqrt{5} - 2)$
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C) $3(\sqrt{5} - 1)$

cracku

cracku

14. The number of coins collected per week by two coin-collectors A and B are in the ratio 3 : 4. If the total number of coins collected by A in 5 weeks is a multiple of 7, and the total number of coins collected by B in 3 weeks is a multiple of 24, then the minimum possible number of coins collected by A in one week is

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Answer: _____

Handwritten notes:

- $3x$ (A), $4x$ (B)
- $5 \times 3x = 15x$ (multiple of 7)
- $3 \times 4x = 12x$ (multiple of 24)
- $x = 7y$
- $21y \times 5 = 105y$

1 week

15x → multiple of 7

12x → multiple of 24

14

21

28

15. A fruit seller has a stock of mangoes, bananas and apples with at least one fruit of each type. At the beginning of a day, the number of mangoes make up 40% of his stock. That day, he sells half of the mangoes, 96 bananas and 40% of the apples. At the end of the day, he ends up selling 50% of the fruits. The smallest possible total number of fruits in the stock at the beginning of the day is

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Answer: _____

Handwritten notes:

- Mangoes: $0.4N$
- Bananas: 96
- Apples: $0.6N$
- Sold: $0.2N$ (mangoes), 96 (bananas), $0.4A$ (apples)
- $0.4N \rightarrow \frac{2N}{5}$
- $N \rightarrow \text{multiple of } 5$
- $A \rightarrow \text{multiple of } 5$

16. Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is

cracku

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Answer: _____

Handwritten notes:

$$G + S = 0$$

1 day
Gautam = G
Suhani = S
work = 20(G+S)
20(0.6G + 1.5S)

0.6G
1.5S



VIDEO SOLUTION

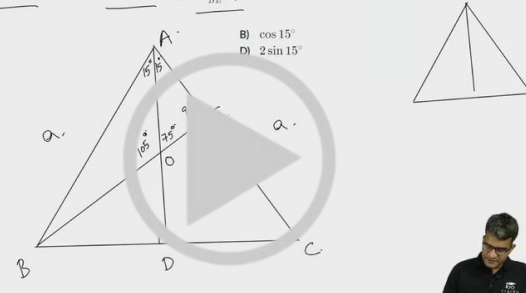
17. Let $\triangle ABC$ be an isosceles triangle such that AB and AC are of equal length. AD is the altitude from A on BC and BE is the altitude from B on AC. If AD and BE intersect at O such that $\angle AOB = 105^\circ$, then $\frac{AD}{BE}$ equals

- A $\sin 15^\circ$
- B $\cos 15^\circ$
- C $2 \cos 15^\circ$
- D $2 \sin 15^\circ$

cracku

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A) $\sin 15^\circ$
B) $\cos 15^\circ$
C) $2 \cos 15^\circ$
D) $2 \sin 15^\circ$




VIDEO SOLUTION

18. A rectangle with the largest possible area is drawn inside a semicircle of radius 2 cm. Then, the ratio of the lengths of the largest to the smallest side of this rectangle is

- A 2 : 1
- B 1 : 1
- C $\sqrt{5} : 1$
- D $\sqrt{2} : 1$

A rectangle with the largest possible area is drawn inside a semicircle of radius 2 cm. Then, the ratio of the lengths of the largest to the smallest side of this rectangle is

A) 2:1
B) 1:1
C) $\sqrt{5}:1$
D) $\sqrt{2}:1$

cracku

VIDEO SOLUTION

19. In a regular polygon, any interior angle exceeds the exterior angle by 120 degrees. Then, the number of diagonals of this polygon is

In a regular polygon, any interior angle exceeds the exterior angle by 120 degrees. Then, the number of diagonals of this polygon is

Answer: _____

N sided regular.

exterior = $\frac{360}{N}$

interior = $\frac{(N-2)180}{N}$

$\frac{360}{N}$

$(N-2)180$

$(N-2)180$

360

$(N-2)180$

$(N-2)180$

cracku

VIDEO SOLUTION

20. The value of $1 + \left(1 + \frac{1}{3}\right) \frac{1}{4} + \left(1 + \frac{1}{3} + \frac{1}{9}\right) \frac{1}{16} + \left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27}\right) \frac{1}{64} + \dots$ is

- A $\frac{15}{13}$
- B $\frac{27}{12}$
- C $\frac{15}{8}$
- D $\frac{16}{11}$

The value of $\left(1 + \frac{1}{4}\right) + \left(1 + \frac{1}{16}\right) + \left(1 + \frac{1}{64}\right) + \left(1 + \frac{1}{256}\right) + \dots$ is

A) $\frac{27}{16}$
B) $\frac{27}{12}$
C) $\frac{27}{11}$
D) $\frac{27}{10}$

$a, ar, ar^2, \dots \infty$
 $-1 < r < 1$
 $\frac{a}{1-r}$

$1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$
 $= \frac{1}{1 - \frac{1}{4}}$
 $= \frac{4}{3}$
 $\frac{1}{3} \left[\frac{4}{3} + \frac{4}{3} + \frac{4}{3} + \dots \right]$
 $= \frac{1}{3} \times \frac{4}{3} \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \right]$




VIDEO SOLUTION

21. Let $a_n = 46 + 8n$ and $b_n = 98 + 4n$ be two sequences for natural numbers $n \leq 100$. Then, the sum of all terms common to both the sequences is

- A 14900
B 14798
C 15000
D 14602

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A) 14900
B) 14798
C) 15000
D) 14602

54, 62, 70, 78, 86, 94, 102, 110, 118, 126, 134, 142, 150, 158, 166, 174, 182, 190, 198, 206, 214, 222, 230, 238, 246, 254, 262, 270, 278, 286, 294, 302, 310, 318, 326, 334, 342, 350, 358, 366, 374, 382, 390, 398, 406, 414, 422, 430, 438, 446, 454, 462, 470, 478, 486, 494, 502, 510, 518, 526, 534, 542, 550, 558, 566, 574, 582, 590, 598, 606, 614, 622, 630, 638, 646, 654, 662, 670, 678, 686, 694, 702, 710, 718, 726, 734, 742, 750, 758, 766, 774, 782, 790, 798, 806, 814, 822, 830, 838, 846, 854, 862, 870, 878, 886, 894, 902, 910, 918, 926, 934, 942, 950, 958, 966, 974, 982, 990, 998, 1006, 1014, 1022, 1030, 1038, 1046, 1054, 1062, 1070, 1078, 1086, 1094, 1102, 1110, 1118, 1126, 1134, 1142, 1150, 1158, 1166, 1174, 1182, 1190, 1198, 1206, 1214, 1222, 1230, 1238, 1246, 1254, 1262, 1270, 1278, 1286, 1294, 1302, 1310, 1318, 1326, 1334, 1342, 1350, 1358, 1366, 1374, 1382, 1390, 1398, 1406, 1414, 1422, 1430, 1438, 1446, 1454, 1462, 1470, 1478, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, 1630, 1638, 1646, 1654, 1662, 1670, 1678, 1686, 1694, 1702, 1710, 1718, 1726, 1734, 1742, 1750, 1758, 1766, 1774, 1782, 1790, 1798, 1806, 1814, 1822, 1830, 1838, 1846, 1854, 1862, 1870, 1878, 1886, 1894, 1902, 1910, 1918, 1926, 1934, 1942, 1950, 1958, 1966, 1974, 1982, 1990, 1998, 2006, 2014, 2022, 2030, 2038, 2046, 2054, 2062, 2070, 2078, 2086, 2094, 2102, 2110, 2118, 2126, 2134, 2142, 2150, 2158, 2166, 2174, 2182, 2190, 2198, 2206, 2214, 2222, 2230, 2238, 2246, 2254, 2262, 2270, 2278, 2286, 2294, 2302, 2310, 2318, 2326, 2334, 2342, 2350, 2358, 2366, 2374, 2382, 2390, 2398, 2406, 2414, 2422, 2430, 2438, 2446, 2454, 2462, 2470, 2478, 2486, 2494, 2502, 2510, 2518, 2526, 2534, 2542, 2550, 2558, 2566, 2574, 2582, 2590, 2598, 2606, 2614, 2622, 2630, 2638, 2646, 2654, 2662, 2670, 2678, 2686, 2694, 2702, 2710, 2718, 2726, 2734, 2742, 2750, 2758, 2766, 2774, 2782, 2790, 2798, 2806, 2814, 2822, 2830, 2838, 2846, 2854, 2862, 2870, 2878, 2886, 2894, 2902, 2910, 2918, 2926, 2934, 2942, 2950, 2958, 2966, 2974, 2982, 2990, 2998, 3006, 3014, 3022, 3030, 3038, 3046, 3054, 3062, 3070, 3078, 3086, 3094, 3102, 3110, 3118, 3126, 3134, 3142, 3150, 3158, 3166, 3174, 3182, 3190, 3198, 3206, 3214, 3222, 3230, 3238, 3246, 3254, 3262, 3270, 3278, 3286, 3294, 3302, 3310, 3318, 3326, 3334, 3342, 3350, 3358, 3366, 3374, 3382, 3390, 3398, 3406, 3414, 3422, 3430, 3438, 3446, 3454, 3462, 3470, 3478, 3486, 3494, 3502, 3510, 3518, 3526, 3534, 3542, 3550, 3558, 3566, 3574, 3582, 3590, 3598, 3606, 3614, 3622, 3630, 3638, 3646, 3654, 3662, 3670, 3678, 3686, 3694, 3702, 3710, 3718, 3726, 3734, 3742, 3750, 3758, 3766, 3774, 3782, 3790, 3798, 3806, 3814, 3822, 3830, 3838, 3846, 3854, 3862, 3870, 3878, 3886, 3894, 3902, 3910, 3918, 3926, 3934, 3942, 3950, 3958, 3966, 3974, 3982, 3990, 3998, 4006, 4014, 4022, 4030, 4038, 4046, 4054, 4062, 4070, 4078, 4086, 4094, 4102, 4110, 4118, 4126, 4134, 4142, 4150, 4158, 4166, 4174, 4182, 4190, 4198, 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5534, 5542, 5550, 5558, 5566, 5574, 5582, 5590, 5598, 5606, 5614, 5622, 5630, 5638, 5646, 5654, 5662, 5670, 5678, 5686, 5694, 5702, 5710, 5718, 5726, 5734, 5742, 5750, 5758, 5766, 5774, 5782, 5790, 5798, 5806, 5814, 5822, 5830, 5838, 5846, 5854, 5862, 5870, 5878, 5886, 5894, 5902, 5910, 5918, 5926, 5934, 5942, 5950, 5958, 5966, 5974, 5982, 5990, 5998, 6006, 6014, 6022, 6030, 6038, 6046, 6054, 6062, 6070, 6078, 6086, 6094, 6102, 6110, 6118, 6126, 6134, 6142, 6150, 6158, 6166, 6174, 6182, 6190, 6198, 6206, 6214, 6222, 6230, 6238, 6246, 6254, 6262, 6270, 6278, 6286, 6294, 6302, 6310, 6318, 6326, 6334, 6342, 6350, 6358, 6366, 6374, 6382, 6390, 6398, 6406, 6414, 6422, 6430, 6438, 6446, 6454, 6462, 6470, 6478, 6486, 6494, 6502, 6510, 6518, 6526, 6534, 6542, 6550, 6558, 6566, 6574, 6582, 6590, 6598, 6606, 6614, 6622, 6630, 6638, 6646, 6654, 6662, 6670, 6678, 6686, 6694, 6702, 6710, 6718, 6726, 6734, 6742, 6750, 6758, 6766, 6774, 6782, 6790, 6798, 6806, 6814, 6822, 6830, 6838, 6846, 6854, 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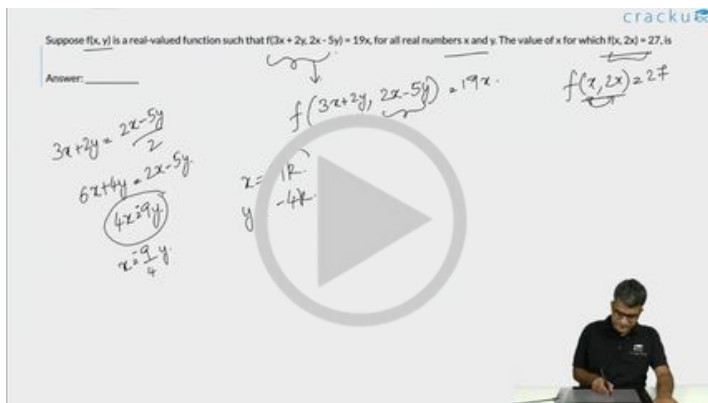
Suppose $f(x, y)$ is a real-valued function such that $f(3x + 2y, 2x - 5y) = 19x$, for all real numbers x and y . The value of x for which $f(x, 2x) = 27$, is

Answer: _____

$3x + 2y = 2x - 5y$
 $6x + 4y = 2x - 5y$
 $4x = -9y$
 $x = -\frac{9}{4}y$

$f(3x + 2y, 2x - 5y) = 19x$
 $f(x, 2x) = 27$

$x = -\frac{12}{4}$
 $y = -4$



VIDEO SOLUTION



Answers

Quant

1.D	2.D	3.A	4.D	5.5	6.468	7.9	8.C
9.D	10.C	11.A	12.C	13.D	14.42	15.340	16.36
17.C	18.A	19.54	20.D	21.A	22.3		

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Explanations

Quant

1. D

It is given that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, which can be written as:

$$\Rightarrow (x^4)^2 + \left(\frac{1}{x^4}\right)^2 = 47$$

$$\Rightarrow (x^4 + \frac{1}{x^4})^2 - 2 \cdot x^4 \cdot \frac{1}{x^4} = 47$$

$$\Rightarrow (x^4 + \frac{1}{x^4})^2 = 49$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 7$$

Similarly, $x^4 + \frac{1}{x^4} = 7$ can be expressed as:

$$\Rightarrow (x^2)^2 + \left(\frac{1}{x^2}\right)^2 = 7$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 - 2 \cdot x^2 \cdot \frac{1}{x^2} = 7$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 3$$

By the same logic, we get $x + \frac{1}{x} = \sqrt{5}$

Now, $x^3 + \frac{1}{x^3} = (x + \frac{1}{x})^3 - 3 \cdot x \cdot \frac{1}{x} (x + \frac{1}{x})$

$$\Rightarrow x^3 + \frac{1}{x^3} = (\sqrt{5})^3 - 3\sqrt{5} = 2\sqrt{5}$$

By the same logic, we can say that

$$\Rightarrow x^9 + \frac{1}{x^9} = (x^3 + \frac{1}{x^3})^3 - 3 \cdot x^3 \cdot \frac{1}{x^3} (x^3 + \frac{1}{x^3})$$

$$\Rightarrow x^9 + \frac{1}{x^9} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$\Rightarrow x^9 + \frac{1}{x^9} = 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

The correct option is D

 VIDEO SOLUTION

2. D

It is given that there are exactly 41 numbers, which can be expressed as the power of two, and exist between 8^m and 8^n , (where m, and n are positive integers, and $m < n$)

$$\text{Hence, } 2^{3m} < 41 \text{ numbers} < 2^{3n}$$

Since, m is a positive integer, the least value of m is 1. Therefore, $2^{3m} = 2^3$, hence, the 41 numbers between them are $2^4, 2^5, 2^6, \dots, 2^{44}$.

Then the lowest possible value of 8^n is 2^{45} . Hence, the smallest value of n is $2^{45} = 8^n \Rightarrow 2^{3n} = 2^{45} \Rightarrow n = 15$

Hence, the smallest value of $m+n$ is $(15+1) = 16$

The correct option is D

 VIDEO SOLUTION

3. A

It is given that for some real numbers a and b , the system of equations $x + y = 4$ and $(a + 5)x + (b^2 - 15)y = 8b$ has infinitely many solutions for x and y .

Hence, we can say that

$$\Rightarrow \frac{a+5}{1} = \frac{b^2-15}{1} = \frac{8b}{4}$$

This equation can be used to find the value of a , and b .

Firstly, we will determine the value of b .

$$\Rightarrow \frac{b^2-15}{1} = \frac{8b}{4} \Rightarrow b^2 - 2b - 15 = 0$$

$$\Rightarrow (b - 5)(b + 3) = 0$$

Hence, the values of b are 5, and -3, respectively.

The value of a can be expressed in terms of b , which is $a + 5 = b^2 - 15 \Rightarrow a = b^2 - 20$

$$\text{When } b = 5, a = 5^2 - 20 = 5$$

$$\text{When } b = -3, a = 3^2 - 20 = -11$$

The maximum value of $ab = (-3) \cdot (-11) = 33$

The correct option is A

 VIDEO SOLUTION

4. D

It is given that $\frac{1}{2}$, $\frac{\log_3(2^x-9)}{\log_3 4}$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ are in an arithmetic progression.

$\frac{\log_3(2^x-9)}{\log_3 4}$ can be written as $\log_4(2^x - 9)$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ can be written as $\log_4(2^x + \frac{17}{2})$

$$\text{Hence, } 2 \log_4(2^x - 9) = \frac{1}{2} + \log_4(2^x + \frac{17}{2})$$

$\frac{1}{2}$ can be written as $\log_4 2$.

Therefore,

$$\Rightarrow 2 \log_4(2^x - 9) = \log_4 2 + \log_4(2^x + \frac{17}{2})$$

$$\Rightarrow \log_4(2^x - 9)^2 = \log_4 2(2^x + \frac{17}{2})$$

$$\Rightarrow (2^x - 9)^2 = 2(2^x + \frac{17}{2})$$

$$\Rightarrow 2^{2x} - 18 \cdot 2^x + 81 = 2 \cdot 2^x + 17$$

$$\Rightarrow 2^{2x} - 20 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^{2x} - 16 \cdot 2^x - 4 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^x(2^x - 16) - 4(2^x - 16) = 0$$

$$\Rightarrow (2^x - 4)(2^x - 16) = 0$$

The values of 2^x can't be 4 (log will be undefined), which implies The value of 2^x is 16.

Therefore, the common difference is $\log_4(2^x - 9) - \log_4 2$

$$\Rightarrow \log_4 7 - \log_4 2 = \log_4\left(\frac{7}{2}\right)$$

The correct option is D

 VIDEO SOLUTION

5.5

It is given that $5^{n-1} < 3^{n+1}$, where n is a natural number. By inspection, we can say that the inequality holds when $n = 1, 2, 3, 4$, and 5 .

Now, we need to find the least integer value of m that satisfies $3^{n+1} < 2^{n+m}$

For, $n = 1$, the least integer value of m is 3 .

For, $n = 2$, the least integer value of m is 3

For, $n = 3$, the least integer value of m is 4 .

For, $n = 4$, the least integer value of m is 4 .

For, $n = 5$, the least integer value of m is 5 .

Hence, the least integer value of m such that for all the values of n , the equation holds is 5.3

▶ VIDEO SOLUTION

6.468

We know that the number of factors of these two numbers is 15 . We know that the factors of 15 are $1, 3, 5$, and 15 .

The number of factors of N is $(p+1) \cdot (q+1)$ (Where, $N = a^p \cdot b^q$, and a, b are prime numbers).

Hence, the value of N will be least when $(p+1)$ and $(q+1)$ are as close as possible and a , and b are the least distinct prime numbers.

Hence, $p+1 = 3 \Rightarrow p = 2$, and $q+1 = 5 \Rightarrow q = 4$, and the prime numbers a , and b are 2 , and 3 , respectively.

Hence, the lowest value of N is $N = 2^4 \times 3^2 = 144$, and the second lowest value of N is $N = 2^2 \times 3^4 = 324$.

Hence, the sum is $(144+324) = 468$

▶ VIDEO SOLUTION

7.9

It is given that $x^2 + bx + c = 0$ has two real roots. Let the roots of the equation be α, β ($\alpha > \beta$)

Then, we can say that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$ Eq(1)

Similarly, $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{5}{9}$ Eq (2)

Eq(2) can be written as $\left(\frac{1}{\alpha} - \frac{1}{\beta}\right)^2 + 2 \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{5}{9}$

$$\Rightarrow \left(\frac{1}{3}\right)^2 + 2 \cdot \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{5}{9}$$

$$\Rightarrow \frac{2}{\alpha \cdot \beta} = \frac{4}{9} \Rightarrow \frac{1}{\alpha \cdot \beta} = \frac{2}{9}$$

$$\Rightarrow \alpha \cdot \beta = \frac{9}{2}$$

We know that the product of the roots is equal to c , which implies $c = \frac{9}{2}$

We also know that the sum of the roots is equal to $-b$.

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^2 - \frac{2}{\alpha \beta} = \frac{5}{9}$$

$$\Rightarrow \left(\frac{\alpha + \beta}{\alpha \beta}\right)^2 - \frac{4}{9} = \frac{5}{9}$$

$$\Rightarrow \left(\frac{\alpha + \beta}{\alpha \beta}\right)^2 = (1)^2$$

$$\Rightarrow \alpha + \beta = \pm \alpha \beta$$

Hence, the maximum value of b is $\frac{9}{2}$.

Hence, the maximum value of $(b+c)$ is 9

 VIDEO SOLUTION

8. C

It is given that a merchant purchases a cloth at a rate of Rs.100 per meter and receives 5 cm length of cloth free for every 100 cm length of cloth purchased by him.

Hence, the cost price of 105 cm clothes is 100 rupees.

It is also known that he marked the price of 100 cm clothes as 110 rupees, and gave a 5% discount, and he cheated his customers by giving 95 cm length of cloth for every 100 cm length of cloth purchased by the customers.

Hence, the selling price of 95 cm clothes is $110 \times (19/20)$ rupees.

Therefore, the selling price of 105 cm clothes is 115.5 rupees.

Hence, the profit is 15.5%

The correct option is C

 VIDEO SOLUTION

9. D

Let the work done by Rahul, Rakshita, and Gurmeet be a , b , and c units per day, respectively, and the total units of work are W .

Hence, we can say that $7(a+b+c) < W$ (Rahul, Rakshita, and Gurmeet, working together, would have taken more than 7 days to finish a job).

Similarly, we can say that $15(a+c) > W$ (Rahul and Gurmeet, working together would have taken less than 15 days to finish the job)

Now, comparing these two inequalities, we get: $7(a+b+c) < W < 15(a+c)$

It is also known that they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. Therefore, the total units of work done is: $W = 6(a+b+c) + 3b$

Hence, we can say that $7(a+b+c) < 6(a+b+c) + 3b < 15(a+c)$

Therefore, $(a+b+c) < 3b \Rightarrow a+c < 2b$, and $9b < 9(a+c) \Rightarrow b < a+c$

$\Rightarrow a+b+c < 3b \Rightarrow 7(a+b+c) < 21b$, and $15b < 15(a+c)$

Hence, The number of days required for b must be in between 15 and 21 (both exclusive).

Hence, the correct option is D

 VIDEO SOLUTION

10. C

It is given that the population of the town in 2020 was 100000. The population decreased by $y\%$ from the year 2020 to 2021 and increased by $x\%$ from the year 2021 to 2022, where x and y are two natural numbers.

Hence, the population in 2021 is $100000 \left(\frac{100-y}{100} \right)$.

The population in 2022 is $100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right)$

It is also given that the population in 2022 was greater than the population in 2020 and the difference between x and y is 10.

Hence,

$$100000 \left(\frac{100-y}{100} \right) \left(\frac{100+x}{100} \right) > 100000, \text{ and } (x-y) = 10$$

$$\Rightarrow 100000 \left(\frac{100-y}{100} \right) \left(\frac{110+y}{100} \right) > 100000$$

$$\Rightarrow \frac{100-y}{100} \left(\frac{110+y}{100} \right) > 1$$

To get the minimum possible value of 2021, we need to increase the value of y as much as possible.

$$\text{Hence, } (100 - y) \{ (100 + y) + 10 \} > 10000$$

$$\Rightarrow 10000 - y^2 + 1000 - 10y > 10000$$

$$\Rightarrow y^2 + 10y < 1000$$

$$\Rightarrow y^2 + 10y + 25 < 1025$$

$$\Rightarrow (y + 5)^2 = 1024 < 1025$$

$$\Rightarrow (y + 5)^2 = 32^2$$

$$\Rightarrow y = 27$$

Hence, the population in 2021 is $100000 \times (100 - 27) = 73000$

The correct option is C

 VIDEO SOLUTION

11. A

Let the volume of mixture A be 200 ml, which implies the quantity of cocoa in the mixture is 120 ml, and the quantity of sugar in the mixture 80 ml.

Similarly, let the volume of the mixture be 300 ml, which implies the quantity of coffee, and sugar in the mixture is 210, and 90 ml, respectively.

Now we combine mixture A, and B in the ratio of 2:3 (if 200 ml mixture A, then 300 ml of mixture B).

Hence, the volume of the mixture C is $(200 + 300) = 500$ ml, and the quantity of the sugar is $(90 + 80) = 170$ ml.

Now he mixes C with an equal amount of milk to make a drink, which implies the quantity of the final mixture is $(500 + 500) = 1000$ ml.

The quantity of sugar in the final mixture is 170 ml.

Hence, the percentage is 17%

The correct option is A

 VIDEO SOLUTION

12. C

Let us assume that A, B, C, D, and E weights are a, b, c, d, and e.

1st condition

$$\frac{(a+b+c)}{3} - \frac{(a+b+c+d)}{4} = x$$

2nd condition

$$\frac{(a+b+c+e)}{4} - \frac{(a+b+c)}{3} = 2x$$

Adding both the equations, we get:

$$\frac{(e-d)}{4} = 3x$$

$$\Rightarrow \frac{(e-d)}{4} = 3x \Rightarrow e - d = 12x$$

Given that $12x = 12 \Rightarrow x = 1$.

13.D

Let us assume the speed of the 1st boat is b , the 2nd boat is s , and the river's speed is r .

Let ' d ' be the distance between A and B.

$$\Rightarrow d = 2(b+r) \text{ and } d = 3(b-r)$$

$$\Rightarrow b + r = d/2 \text{ and } b - r = d/3 \Rightarrow r = d/12 \text{ (subtracting both equations).}$$

Now, it is given that

$$\frac{d}{s+r} + \frac{d}{s-r} = 6$$

$$\Rightarrow \frac{d}{s + \frac{d}{12}} + \frac{d}{s - \frac{d}{12}} = 6$$

$$\Rightarrow 2ds = 6 \left(s^2 - \frac{d^2}{144} \right)$$

$$\Rightarrow 144s^2 - 48ds - d^2 = 0$$

Solving the quadratic equation, we get:

$$s = d \left(\frac{(48 + \sqrt{48^2 + 4(144)})}{2 \times 144} \right)$$

$$s = d \left(\frac{1}{6} + \frac{\sqrt{5}}{12} \right)$$

$$\Rightarrow \text{Required value of } \frac{d}{s+r}$$

$$= \frac{d}{\frac{d}{6} + \frac{\sqrt{5}d}{12} + \frac{d}{12}}$$

$$= \frac{12}{3 + \sqrt{5}} = \frac{(12)(3 - \sqrt{5})}{4}$$

$$= 3(3 - \sqrt{5})$$

14.42

Given that the number of coins collected per week by two coin-collectors, A and B, are in the ratio 3: 4

Let us assume A collects $3c$ coins per week and B collects $4c$ coins per week.

Total number of coins collected by A in 5 weeks = $5 \times 3c = 15c$, which should be multiple of 7 $\Rightarrow c$ should be multiple of 7.

Total number of coins collected by B in 3 weeks = $3 \times 4c = 12c$, which should be a multiple of 24 $\Rightarrow c$ should be a multiple of 2.

So, the least possible value of c is $\text{lcm}(2,7) = 14$.

Coins sold by A in a week = $3c = 3 \times 14 = 42$.

15.340

Let us assume the initial stock of all the fruits is S.

Let us take we have 'b' and 'a' mangoes initially.

Stock of Mangoes = 40% of S = $2S/5$

The total number of fruits sold are Mangoes Sold + Apples Sold + Bananas Sold

$$= 2S/10 + 96 + 4a/10 = S/2 \text{ (Given)}$$

$$\Rightarrow S/5 + 96 + 2a/5 = S/2$$

$$\Rightarrow S = \frac{(4a + 960)}{3}$$

$$\Rightarrow \frac{4a}{3} + 320$$

'a' has to be a multiple of 3 for the above term to be an integer.

But 'a' has to be a multiple of 5 for $4a/10$ to be an integer.

$\Rightarrow$ The smallest value of 'a' satisfying both conditions is 15.

$$\Rightarrow \frac{4a}{3} + 320 = \frac{4(15)}{3} + 320 = 340$$

 VIDEO SOLUTION

16.36

Let 'g' and 's' be the efficiencies of Gautam and Suhani. Let W is the total amount of work.

$$\Rightarrow g + s = W/20 \text{ (1 day work) } \text{---(1)}$$

Also Gautam doing only 60% $\Rightarrow 3g/5$ and Suhani doing 150% $\Rightarrow 3s/2$

$$\Rightarrow 3g/5 + 3s/2 = W/20 \text{ (1 day work)}$$

$$\Rightarrow g + s = \frac{3g}{5} + \frac{3s}{2}$$

$$\Rightarrow \frac{s}{g} = \frac{4}{5} \Rightarrow \text{Gautam is the more efficient person.}$$

Now, from the 1st equation

$$\Rightarrow g + \frac{4g}{5} = \frac{W}{20}$$

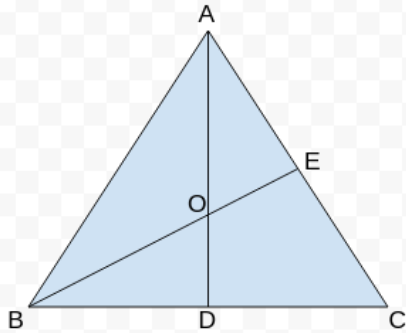
$$\Rightarrow \frac{9}{5}g = \frac{W}{20}$$

$$\Rightarrow g = \frac{W}{36}$$

$\Rightarrow$ Gautam takes 36 days to finish the complete work.

 VIDEO SOLUTION

17.C



Given that $AB = AC \Rightarrow \text{Angle } C = \text{Angle } B$ (1)

AD and BE are altitudes $\Rightarrow$ they make 90 degrees with the sides

$\text{Angle } AOB = 105 \Rightarrow \text{Angle } EOD = 105$ (Vertically Opposite Angles)

In quadrilateral $DOEC$

$\text{Angle } C = 360 - 105 - 90 - 90 = 75 \Rightarrow \text{Angle } B = 75$ (from 1)

We know that from the area of the triangle $AD * BC = BE * AC$

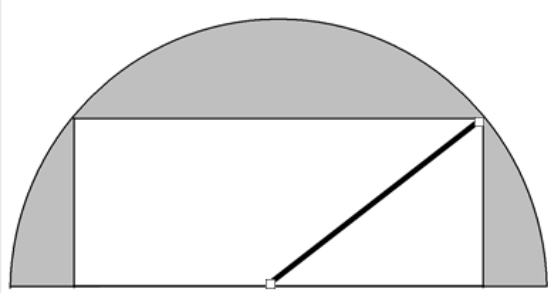
$$\Rightarrow \frac{AD}{BE} = \frac{AC}{BC} = \frac{2R \sin B}{2R \sin A} = \frac{\sin(75)}{\sin(30)} = 2 \sin(75) = 2 \cos(15)$$

$[\sin(x) = \cos(90-x)]$

[VIDEO SOLUTION](#)

18.A

Let us assume the length of the rectangle is ' l ' and breadth of the rectangle is ' b '.



The radius, $l/2$ and b in the above diagram form a right-angled triangle.

$$\Rightarrow \left(\frac{l}{2}\right)^2 + b^2 = 2^2$$

We know that the area of the rectangle is $l*b$, which can be obtained by considering 2 times the geometric mean of $\left(\frac{l}{2}\right)^2$ and b^2 .

Therefore, for the maximum area, the equality condition of AM-GM inequality should be satisfied

$$\Rightarrow \left(\frac{l}{2}\right)^2 = b^2 \Rightarrow l = 2b.$$

$$\Rightarrow l/b = 2/1.$$

[VIDEO SOLUTION](#)

19.54

The sum of the interior angles of a polygon of ' n ' sides is given by $(2n - 4) \times 90$, and the sum of the exterior angles of a polygon is 360 degrees.

So, the difference between them will be $120 * n$

$$\Rightarrow (2n - 4) 90 - 360 = 120n$$

$$\Rightarrow 60n = 720 \Rightarrow n = 12.$$

We know that the number of diagonals of a regular polygon is $nC2 - n = 12C2 - 12 = 66 - 12 = 54$.

 VIDEO SOLUTION

20. D

The given sequence can be written as:

$$1 \left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right) + \frac{1}{3} \left(\frac{1}{4} + \frac{1}{16} + \dots \right) + \frac{1}{9} \left(\frac{1}{16} + \frac{1}{64} + \dots \right) + \dots$$

We know that the sum of an infinite G.P. is $\frac{a}{1-r}$, where a is the first term and r is the common ratio.

$$\Rightarrow \text{The first term} = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$$

$$\Rightarrow \text{The second term} = \frac{1}{3} \left(\frac{\left(\frac{1}{4}\right)}{1-\left(\frac{1}{4}\right)} \right) = \frac{1}{9}$$

$$\Rightarrow \text{The third term} = \frac{1}{9} \left(\frac{\left(\frac{1}{16}\right)}{1-\left(\frac{1}{4}\right)} \right) = \frac{1}{108}$$

Observing these three terms, we see that they are in G.P. with a common ratio of $\frac{1}{12}$

$$\Rightarrow \text{Sum of this infinite G.P.} = \frac{\left(\frac{4}{3}\right)}{1 - \left(\frac{1}{12}\right)} = \frac{16}{11}$$

 VIDEO SOLUTION

21. A

The first series goes as follows:

46, 54, 62, 70, 78, 86, 94, 102,

The second series goes as follows:

98, 102, 106, 110, ...

The first common term is 102 (first term of the common terms) and the common difference between them will be $\text{hcf}(4, 8) = 8$

$\Rightarrow$ The required sequence is 102, 110, 118, (last term should be less than 468 (100th term of second series))

$$\Rightarrow 102 + (n-1)(8) \leq 498$$

$$\Rightarrow n \text{ is less than or equal to } 50.5 \Rightarrow n = 50$$

Using the summation of A.P. formula:

$$\text{Required sum} = \frac{n}{2} (2a + (n-1)d) = \frac{50}{2} (2 \times 102 + 49 \times 8) = 14900$$

 VIDEO SOLUTION

22. 3

Given that $f(3x + 2y, 2x - 5y) = 19x$.

Let us assume the function $f(a, b)$ is a linear combination of a and b .

$$\Rightarrow f(3x+2y, 2x-5y) = m(3x+2y) + n(2x-5y) = 19x$$

$$\Rightarrow 3m + 2n = 19 \text{ and } 2m - 5n = 0$$

Solving we get $m = 5$ and $n = 2$

$$\Rightarrow f(a,b) = 5a+2b$$

$$\Rightarrow f(x,2x) = 5x + 2(2x) = 9x = 27 \Rightarrow x = 3.$$

 VIDEO SOLUTION